# Sina Hosseini

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## Research Profile

Chemical engineer with research and industry experience in heat transfer, CFD, and thermal insulation, combining computational work (CFD, scientific Python) with experimental work (aerogel synthesis, insulation testing, process chemistry). Two journal articles, two conference papers, and three manuscripts in preparation.

**Research interests:** Heat transfer enhancement and optimization; computational fluid dynamics (CFD) and numerical simulation of thermal-fluid systems; thermal insulation, aerogels, and porous media for energy and building applications; thermal management of electronics and data-center cooling; thermal energy storage, including cryogenic storage for hydrogen and LNG

## Education

**Chemistry and Chemical Engineering Research Center of Iran,** Tehran, Iran

- A national research institute for chemistry and chemical engineering in Iran (est. 1989)
- Master of Science in Chemical Engineering, September 2022 – June 2026
- Thesis: Production of Magnesium Oxide from Magnesite Ore, Advisor: Dr. Reza Panahi
- GPA 16.16/20 (3.28/4.0, converted with Scholaro)

**University of Zanjan,** Zanjan, Iran

- Bachelor of Science in Chemical Engineering, September 2018 – September 2022
- Thesis: Investigation of nanoporous silica aerogel additive to construction plaster, Advisor: Dr. Taher Yousefi Amiri
- GPA 17.31/20 (3.60/4.0, converted with Scholaro)

**Shahid Soltani 3 High School,** National Organization for Development of Exceptional Talents (NODET)

- High School Diploma in Mathematics and Physics, September 2018
- GPA 3.444/4.0

## Experience

**Nano Sun Abrad Co.,** Abyek, Qazvin Province, Iran

Head of Research & Development, January 2023 – September 2025

- Led R&D for vacuum-insulated cryogenic storage tanks as the only engineer in the R&D unit, supervising four engineers from production
- Reduced thermal bridges at supports, nozzles, bolts, and seams, guided by thermal imaging and FloEFD models
- Validated the models against boil-off measurements and measured temperatures; simulated temperatures agreed within 5 %

**MAPNA Turbine Engineering and Manufacturing Co. (TUGA),** Fardis, Alborz Province, Iran

Chemical Engineer Intern, August 2022 – November 2022

- Conducted quality and condition testing of industrial lubricating oils and coolants in the Machinery Test & Control Department
- Performed calibration and quality checks on precision machining equipment

**Pakan Atiye Nano Danesh Co. (PANDA),** Qazvin, Iran

Junior Engineer, November 2021 – August 2022

- Ran FloEFD heat-transfer simulation cases for a battery-powered ladle carrier at Mobarakeh Steel Co.
- Produced 3D component and assembly designs using CAD software (CATIA, SolidWorks)
- Co-drafted the insulation and cooling recommendations, which were implemented on the vehicle

**Ahmadi Roshan Research Project,** Iran's National Elites Foundation

Python Developer, September 2020 – November 2022

- Wrote Python modules for a Darcy-flow reservoir simulator, including streamline visualization and results display
- Mentored three to five junior team members in Python programming and numerical methods

**Nanotechnology Research Laboratory,** University of Zanjan, Zanjan, Iran

Research Assistant, July 2019 – August 2022

- Conducted experimental research on nanoporous materials, contributing to two peer-reviewed publications
- Operated and maintained laboratory characterization equipment and research instrumentation
- Applied research techniques in sample preparation, testing, and data analysis

**Faculty of Engineering,** University of Zanjan, Zanjan, Iran

Teaching Assistant, February 2021 – January 2022

- Served as teaching assistant for core chemical-engineering courses: Material & Energy Balances, Heat Transfer, Thermodynamics II, and Engineering Mathematics
- Led problem-solving sessions and supported students in improving their academic performance

## Publications

- H. Bargozin, T. Yousefi Amiri, and S. Hosseini, "Reduce building energy loss with liquid thermal insulation containing nanoporous silica aerogels," Journal of Energy Management and Technology, vol. 7, no. 1, pp. 11–18, 2023.
- M. Maleki Kakelar, S. Hosseini, and A. Mohammadi, "Feasibility of Using Carbonic Anhydrase in Bio Sequestration of Carbon Dioxide: Challenges and Opportunities," Iranian Chemical Engineering Journal, 2022.
- H. Bargozin, T. Yousefi Amiri, and S. Hosseini, "Reducing building energy loss with liquid thermal insulation containing nanoporous silica aerogel," in 7th International Conference on Technology and Energy Management, 2021.
- H. Bargozin, T. Yousefi Amiri, and S. Hosseini, "Simulation of online replacement of nanoporous aerogel thermal insulation on Asaluyeh LPG lines," in the National Conference of Chemical Engineering and Nanotechnology, 2020.

## Skills & Abilities

**Programming** - Python 3 (NumPy, Matplotlib, VTK), MicroPython, C, MATLAB, and Git
**Simulation & CAD** - FloEFD, STAR-CCM+, CATIA, SolidWorks, Solid Edge, and Proteus
**Other** - Electronics & circuit prototyping, technical writing, project management (Trello), literature and patent searching (EndNote)

## Volunteering Experience

**Journal of Green Processes,** Scientific Association of Chemical Engineering Students

Author, September 2020 – June 2022

- Wrote accessible articles introducing chemical-engineering research and industry topics to a student readership

## Projects

**PicoPhone,** Self-Directed Embedded Systems Project (in progress, 2025 – present)

- Building a mobile phone from scratch on a Raspberry Pi Pico 2 with a SIM800C GSM module
- Wrote the C firmware: a display driver with touch input, and SMS send and receive through the SIM800C (text mode, Unicode and multipart). The parts are tested separately and not yet integrated

## Awards and Honors

- Distinguished Undergraduate Researcher, Faculty of Engineering, University of Zanjan, November 2022
- Ranked 2nd by GPA of 46 students, 2018 entering class, Chemical Engineering, University of Zanjan, September 2022
- 17th place, National Student Olympiad in Chemical Engineering, April 2022
- 26th place, National Student Olympiad in Chemical Engineering, November 2021